



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 31/07/2026

EXPERIMENT 13 -- Simple programming using CASE and LOOP

AIM:

To learn and implement MySQL **CASE** and **LOOP** statements to execute a block of code repeatedly based on a condition.

PROGRAM 1:

Write a function using the **CASE** statement to classify students based on their CGPA:

- CGPA $\leq 7.00 \rightarrow$ **Low**
- CGPA $\leq 9.00 \rightarrow$ **Average**
- CGPA $> 9.00 \rightarrow$ **High**

PROCEDURE:

1. Create a function that accepts the student's register number.
2. Retrieve the student's CGPA from the student_cgpa table.
3. Use the CASE statement to classify the CGPA.
4. Return the corresponding category.
5. Execute the function using the student's register number.

Program

```
DELIMITER //
```

```
CREATE FUNCTION cgpa_category(rno INT)
```

```
RETURNS VARCHAR(20)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    DECLARE student_cgpa_value DECIMAL(3,2);
```

```
    DECLARE category VARCHAR(20);
```

```
    SELECT cgpa
```

```
    INTO student_cgpa_value
```

```
    FROM student_cgpa
```

```
    WHERE reg_no = rno;
```

```
    CASE
```

```
        WHEN student_cgpa_value  $\leq 7.00$  THEN
```

```
            SET category = 'Low';
```

```
        WHEN student_cgpa_value  $\leq 9.00$  THEN
```

```

        SET category = 'Average';

    ELSE
        SET category = 'High';
    END CASE;

    RETURN category;
END //

DELIMITER ;

```

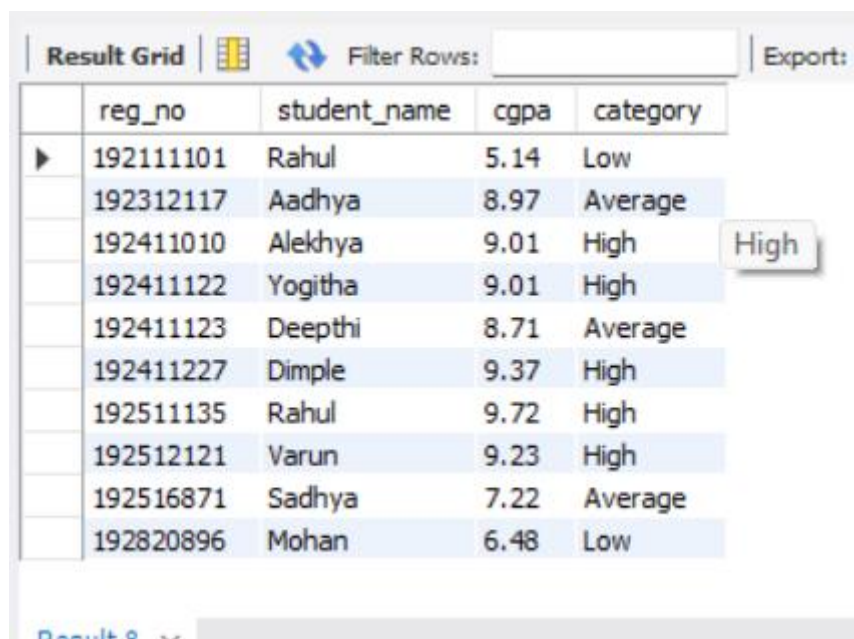
Execute the Function

```

SELECT reg_no, student_name, cgpa,
       cgpa_category(reg_no) AS category
FROM student_cgpa;

```

OUTPUT:



reg_no	student_name	cgpa	category
192111101	Rahul	5.14	Low
192312117	Aadhya	8.97	Average
192411010	Alekhya	9.01	High
192411122	Yogitha	9.01	High
192411123	Deepthi	8.71	Average
192411227	Dimple	9.37	High
192511135	Rahul	9.72	High
192512121	Varun	9.23	High
192516871	Sadhya	7.22	Average
192820896	Mohan	6.48	Low

PROGRAM 2 – LOOP WITH ITERATE

Write a function using **ITERATE** to repeatedly increase a counter while the number of passed subjects is less than 35. Once the value becomes 35 or greater, terminate the LOOP.

PROCEDURE

1. Create a function that accepts the student's register number.
2. Retrieve no_of_subs_pass from student_cgpa.
3. Start a LOOP.
4. If the number of passed subjects is less than 35, increment it.
5. Use ITERATE to repeat the loop.
6. When it reaches 35, use LEAVE to terminate the loop.
7. Return the final value.

PROGRAM

```
DELIMITER //

CREATE FUNCTION pass_subject_loop(rno INT)
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE pass_count INT;

    SELECT no_of_subs_pass
    INTO pass_count
    FROM student_cgpa
    WHERE reg_no = rno;

    pass_loop: LOOP

        IF pass_count < 35 THEN
            SET pass_count = pass_count + 1;
            ITERATE pass_loop;
        END IF;

        LEAVE pass_loop;

    END LOOP pass_loop;

    RETURN pass_count;
END //
```

DELIMITER ;

Execute the Function

```
SELECT reg_no,
       student_name,
       no_of_subs_pass,
       pass_subject_loop(reg_no) AS final_pass_count
FROM student_cgpa;
```

OUTPUT:

Result Grid				
		Filter Rows:		Export:  Wrap Cell Content: 
	reg_no	student_name	no_of_subs_pass	final_pass_count
▶	192111101	Rahul	30	35
	192312117	Aadhya	25	35
	192411010	Alekhya	35	35
	192411122	Yogitha	31	35
	192411123	Deepthi	22	35
	192411227	Dimple	31	35
	192511135	Rahul	37	37
	192512121	Varun	27	35
	192516871	Sadhya	25	35
	192820896	Mohan	18	35

Result 9 ×

RESULT: Thus the Simple programming exercise using CASE and LOOP executed successfully.